Evaluation of the cyto- and genotoxic activity of yerba mate (Ilex paraguariensis) in human lymphocytes in vitro.

Despite its antioxidant capacity and well-known health benefits, yerba mate tea (Ilex paraguariensis) has been shown to possess some genotoxic and mutagenic activities and to increase incidence of some types of cancer. The aim of this study was to estimate the cyto- and genotoxicity of mate tea in human peripheral lymphocytes in vitro. We found that yerba mate extract induced a concentration-dependent, statistically significant increase in the level of apoptotic and necrotic cells and a decrease in the nuclear division index (NDI). Mate-exposed lymphocytes had a reduced transcriptional rDNA activity, which may be due to the stress conditions, and showed an elevated production of micronuclei. The FISH technique revealed the appearance of an acrocentric signal in mate-induced micronuclei, which suggests that under these conditions yerba mate extract may display aneugenic activity. Since caffeine is one of the most abundant compounds found in the dry mass of mate, we conducted additional experiments with caffeine alone. We showed that caffeine used at the same concentrations manifests a more potent cyto- and genotoxic effect that may account, at least in part, for the disadvantageous effects observed for yerba mate extract.